Use Case: Check Game End (Win/Lose/Tie)

Iteration: 1

Primary Actor: System

Goal in Context: Determine if the game has reached an end state after each move, based on win, lose, or tie conditions.

Preconditions:

1. The game is in progress with an active session.
2. A player has completed their move and placed a piece on the board.
3. The system has updated the game state with the latest move.
4. Conditions for winning, losing, or a tie are defined and accessible in the game logic (e.g., three in a row for Tic-Tac-Toe, four in a row for Connect 4, or no moves left in Checkers).

Trigger: A player completes a move.

Scenario:

1. The system updates the board with the player's move.
2. The system evaluates the board state to determine if any end-game conditions are met:
    - **Tic-Tac-Toe**:
        - The system checks for three identical pieces in a row, column, or diagonal to identify a win.
        - The system declares a tie if the board is full and no winning condition is met.
    - **Checkers**:
        - The system checks if one player has no remaining pieces or valid moves, declaring the opponent the winner.
    - **Connect 4**:
        - The system checks for four identical pieces in a row, column, or diagonal to determine a win.
        - The system declares a tie if all columns are filled and no win condition is met.
3. If an end-game condition is met, the system concludes the game and declares the result (win, lose, or tie).
4. The system notifies both players of the game outcome.

Postconditions:

1. If a win, lose, or tie condition is met, the game session is ended and the result is recorded.
2. Else no end-game condition is met and the next turn continues.

Exceptions:

- If an error occurs in evaluating the board state (e.g., system glitch), the system resets the board to its last valid state and prompts the player to attempt the same or new move again.

Priority: High, as identifying game-end conditions is crucial for determining the game's outcome.

When Available: 1st iteration.

Frequency of Use: Once per game at the end of the game.

Channel to Actor: The system automatically verifies players via the game interface.

Secondary Actors: Both players in the game session.

Channels to Secondary Actors: The result is displayed on each player's screen at the end of the game.

Open Issues: None.